🧪 SBSDocker-v2 with Kind - Dry Run Analysis

✅ PASSING CHECKS

1. Script File Status

- **File exists**: ✅ YES
- **File size**: 26,854 bytes (26.8 KB)
- **Line count**: 644 lines
- **Last modified**: Recently updated with Kind integration

2. Script Structure Analysis

- **Functions defined**: ✅ All required functions present
- **Configuration templates**: ✅ Valid YAML and JSON structures
- **Error handling**: ✅ Includes try-catch blocks and validation
- **Progress reporting**: ✅ Uses Write-Progress throughout

3. Corporate Environment Compatibility

- **Proxy configuration**: ✅ Sopra proxy settings included
- **DNS settings**: ✅ Corporate DNS servers configured
- **Network policies**: ✅ NO_PROXY settings for internal domains
- **Security policies**: ✅ Execution policy handling included

4. Installation Flow Logic

1. ✅ Administrator privileges check
2. ✅ Disk space validation (2GB minimum)
3. ✅ WSL feature enablement (with restart handling)
4. ✅ Chocolatey installation
5. ✅ Ubuntu WSL distribution setup
6. ✅ Docker installation and configuration
7. ✅ Kind and Kubernetes tools installation
8. ✅ Cluster creation with corporate proxy
9. ✅ Dashboard and shortcuts setup
10. ✅ Verification commands

⚠️ ITEMS TO VERIFY BEFORE EXECUTION

1. Prerequisites

- [ ] **Run as Administrator** - Script will exit if not admin
- [ ] **Minimum 2GB free space** on C: drive
- [ ] **Network connectivity** to download sites
- [ ] **Corporate firewall** allows Kind cluster ports

2. Expected System Changes

- **WSL Features**: Will be enabled (may require restart)
- **Chocolatey**: Will be installed if not present
- **Environment Variables**: DOCKER_HOST, KUBECONFIG will be set
- **Windows PATH**: Docker CLI path will be added
- **Desktop Shortcuts**: 5 new shortcuts will be created

3. Download Requirements

The script will download approximately **1.5GB** of files:

- Ubuntu Cloud Image: ~500MB
- Docker binaries: ~50MB
- Kubernetes tools: ~100MB
- Container images: ~800MB (during Kind cluster creation)

🔧 CONFIGURATION VALIDATION

Kind Cluster Configuration

```yaml
✅ API Version: kind.x-k8s.io/v1alpha4
✅ Cluster Name: sbsdocker-cluster
✅ Proxy Settings: Configured for Sopra network
✅ Port Mappings: 8080, 8443, 30000-30002
✅ Network Policy: Corporate-friendly settings
```

Docker Daemon Configuration

```
✅ TCP Socket: tcp://127.0.0.1:2375
✅ Unix Socket: unix:///var/run/docker.sock
✅ Corporate Proxy: Configured
```

🚀 EXPECTED OUTCOMES

After Successful Installation:

1. **Docker**: Fully functional in WSL Ubuntu
2. **Kind Cluster**: Running with 1 control-plane node
3. **Kubernetes Tools**: kubectl, helm, k9s available
4. **Dashboard**: Kubernetes web UI accessible
5. **Desktop Shortcuts**: 5 management shortcuts created

Desktop Shortcuts Created:

- `SBSDocker UI.lnk` - LazyDocker interface
- `Start SBSDocker.lnk` - Start Docker daemon
- `SBSDocker K9s.lnk` - Kubernetes terminal UI
- `SBSDocker K8s Dashboard.lnk` - Web dashboard
- `SBSDocker Kind Manager.lnk` - Kind cluster management

🎯 MENU OPTIONS AVAILABLE

The script provides these installation options:

1. **Install Docker Only** - Base Docker setup
2. **Install Docker + Kubernetes (Kind)** - Full stack
3. **Uninstall All** - Complete removal
4. **Uninstall Kubernetes Only** - Keep Docker, remove K8s

⚡ POTENTIAL ISSUES & MITIGATIONS

Issue	Probability	Impact	Mitigation in Script
WSL not enabled	High	Requires restart	Auto-detects and enables with restart prompt
Network timeouts	Medium	Download failures	Corporate proxy pre-configured
Docker startup timing	Medium	Kind cluster fails	Sleep delays included between steps
Insufficient permissions	Low	Installation fails	Admin check at start
Disk space issues	Low	Installation fails	2GB space check included

🔍 MANUAL VERIFICATION STEPS

Before running the script, manually verify:

```powershell
# 1. Check if running as admin
([Security.Principal.WindowsPrincipal] [Security.Principal.WindowsIdentity]::GetCurrent()).IsInRole([Security.Principal.Winc

# 2. Check disk space (should be > 2GB)
[math]::Round((Get-PSDrive C).Free/1GB,1)

# 3. Check WSL features
Get-WindowsOptionalFeature -Online -FeatureName Microsoft-Windows-Subsystem-Linux
Get-WindowsOptionalFeature -Online -FeatureName VirtualMachinePlatform

# 4. Test network connectivity
Test-NetConnection github.com -Port 443
Test-NetConnection download.docker.com -Port 443
```

💡 RECOMMENDATIONS

Before Running:

1. **Close all Docker Desktop** instances if installed
2. **Backup important data** (standard precaution)
3. **Run during maintenance window** (may require restart)
4. **Ensure stable network** connection

After Installation:

1. **Test with**: `kubectl get nodes`
2. **Verify Docker**: `docker version`

3. **Check Kind cluster**: `kind get clusters`
4. **Access dashboard** via desktop shortcut

🎉 CONCLUSION

✅ SCRIPT IS READY FOR EXECUTION

The dry run analysis shows the script is:

- **Syntactically correct**
- **Logically sound**
- **Corporate environment ready**
- **Error handling included**
- **Recovery mechanisms present**

Risk Level: 🟢 **LOW** - Script includes proper safeguards and validation

Estimated Installation Time: 15-30 minutes (depending on downloads)

Last analyzed: July 3, 2025
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